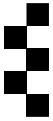
Vector Space Model



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Document Collection Representation

- A collection of *n* documents can be represented in the vector space model by a document-term matrix.
- An entry in the matrix corresponds to the "weight" of a term in the document; zero means the term has no significance in the document or it simply doesn't exist in the document.

$$\begin{pmatrix} & T_1 & T_2 & \ldots & T_t \\ D_1 & w_{11} & w_{21} & \ldots & w_{t1} \\ D_2 & w_{12} & w_{22} & \ldots & w_{t2} \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ D_n & w_{1n} & w_{2n} & \ldots & w_{tn} \end{pmatrix}$$

Both documents and queries are represented using the bag-of-words model.

Vector-Space Model

- *t* distinct terms remain after preprocessing
  - Unique terms that form the VOCABULARY
- These "orthogonal" terms form a vector space.

  Dimension = |vocabulary|

  - 2 terms → bi-dimensional; …; t-terms → t-dimensional
- Each term, *i*, in a document or query *j*, is given a real-valued weight, $w_{ij.}$
- Both documents and queries are expressed as *t*-dimensional vectors:

$$d_j = (w_{1j}, w_{2j}, \ldots, w_{tj})$$

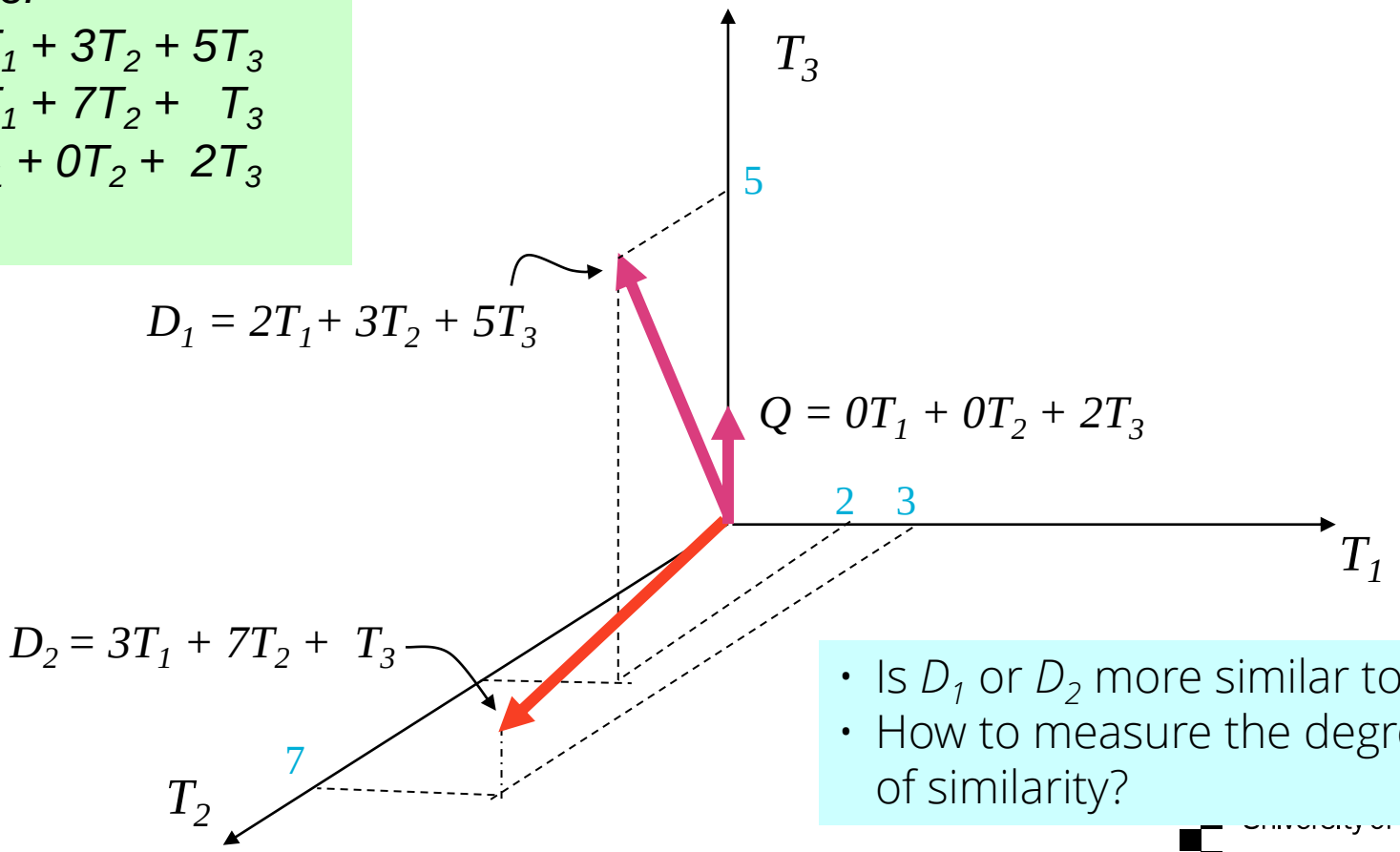
Graphical Representation

Example:

$D_1 = 2T_1 + 3T_2 + 5T_3$

$D_2 = 3T_1 + 7T_2 + T_3$

$Q = 0T_1 + 0T_2 + 2T_3$



- Is D_1 or D_2 more similar to Q?
- How to measure the degree of similarity?

Similarity Measure

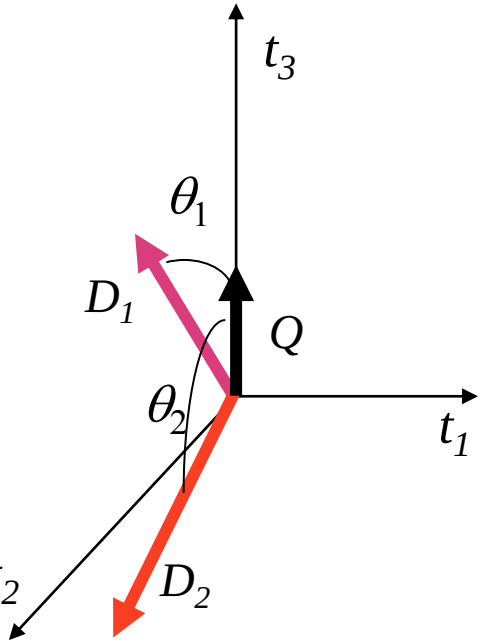
- We now have vectors for all documents in the collection, a vector for the query, how to compute similarity?
- A similarity measure is a function that computes the *degree of similarity* between two vectors.
- Using a similarity measure between the query and each document:
  - It is possible to rank the retrieved documents in the order of presumed relevance.
  - It is possible to enforce a certain threshold so that the size of the retrieved set can be controlled.

Cosine Similarity vs. Inner Product

- Cosine similarity measures the cosine of the angle between two vectors.
- Inner product normalized by the vector lengths.

$$\text{CosSim}(\boldsymbol{d}_j, \boldsymbol{q}) = \frac{\vec{d}_j \cdot \vec{q}}{\left|\vec{d}_j\right| \cdot \left|\vec{q}\right|} = \frac{\sum_{i=1}^{t} (w_{ij} \cdot w_{iq})}{\sqrt{\sum_{i=1}^{t} w_{ij}^2 \cdot \sum_{i=1}^{t} w_{iq}^2}}$$

$$\text{InnerProduct}(\boldsymbol{d}_j, \boldsymbol{q}) = \vec{d}_j \cdot \vec{q}$$



$D_1 = 2T_1 + 3T_2 + 5T_3$ CosSim(D_1 , Q) = 10 / $\sqrt{(4+9+25)(0+0+4)}$ = 0.81
$D_2 = 3T_1 + 7T_2 + 1T_3$ CosSim(D_2 , Q) = 2 / $\sqrt{(9+49+1)(0+0+4)}$ = 0.13
$Q = 0T_1 + 0T_2 + 2T_3$

D_1 is 6 times better than D_2 using cosine similarity but only 5 times better using inner product.

Problems with Vector Space Models (1)

- Simple, mathematically based approach.
- Provides partial matching and ranked results.
- Allows efficient implementation for large document collections.
- Missing semantic information (e.g. word sense).
- Missing syntactic information (e.g. phrase structure, word order, proximity information).

Problems with Vector Space Model (2)

- Lacks the control of a Boolean model (e.g., requiring a term to appear in a document).
  - Given a **two-term query "A B",** may prefer a document containing **A frequently but not B**, over a document that contains **both A and B, but both less frequently**.